

lab-04-solution

```
library(tidyverse)
```

Exercise 1

```
mcd <- read_csv("data/menu.csv")  
  
mcd |>  
  slice(1:7)
```

```
# A tibble: 7 x 6  
  Category Item          Oz Calories  Fat Sugars  
  <chr>    <chr>      <dbl>    <dbl> <dbl> <dbl>  
1 Breakfast Egg McMuffin  4.8      300    13     3  
2 Breakfast Egg White Delight  4.8      250     8     3  
3 Breakfast Sausage McMuffin  3.9      370    23     2  
4 Breakfast Sausage McMuffin with Egg  5.7      450    28     2  
5 Breakfast Sausage McMuffin with Egg Whites  5.7      400    23     2  
6 Breakfast Steak & Egg McMuffin  6.5      430    23     3  
7 Breakfast Bacon, Egg & Cheese Biscuit (Regular Bi~  5.3      460    26     3
```

```
names(mcd)
```

```
[1] "Category" "Item"      "Oz"        "Calories" "Fat"       "Sugars"
```

Exercise 2

```
mcd |>
  count(Category)
```

```
# A tibble: 9 x 2
  Category      n
  <chr>      <int>
1 Beef & Pork    15
2 Beverages     27
3 Breakfast     42
4 Chicken & Fish 27
5 Coffee & Tea   95
6 Desserts       7
7 Salads         6
8 Smoothies & Shakes 28
9 Snacks & Sides 13
```

Most: Coffee & Tea.

Least: Salads.

Exercise 3

a.

```
mcd_cal <- mcd |>
  mutate(cal_oz = Calories/Oz)
```

b.

```
mcd_cal |>
  select(Category, Item, cal_oz) |>
  slice(1:5)
```

```
# A tibble: 5 x 3
  Category Item      cal_oz
  <chr>    <chr>      <dbl>
1 Breakfast Egg McMuffin    62.5
2 Breakfast Egg White Delight 52.1
3 Breakfast Sausage McMuffin 94.9
4 Breakfast Sausage McMuffin with Egg 78.9
5 Breakfast Sausage McMuffin with Egg Whites 70.2
```

Sausage McMuffin has the highest calories per ounce, out of these first five.

Exercise 4

a

```
# a.  
mcd_cal |>  
  select(Item, Calories) |>  
  arrange(desc(Calories)) |>  
  slice(1:5)
```

```
# A tibble: 5 x 2  
  Item                                     Calories  
  <chr>                                <dbl>  
1 Chicken McNuggets (40 piece)          1880  
2 Big Breakfast with Hotcakes (Large Biscuit) 1150  
3 Big Breakfast with Hotcakes (Regular Biscuit) 1090  
4 Big Breakfast with Hotcakes and Egg Whites (Large Biscuit) 1050  
5 Big Breakfast with Hotcakes and Egg Whites (Regular Biscuit) 990
```

b

```
# b.  
mcd_cal |>  
  select(Item, cal_oz) |>  
  arrange(desc(cal_oz)) |>  
  slice(1:5)
```

```
# A tibble: 5 x 2  
  Item                                     cal_oz  
  <chr>                                <dbl>  
1 Chocolate Chip Cookie                160  
2 Oatmeal Raisin Cookie                150  
3 Fat Free Chocolate Milk Jug          130  
4 Cinnamon Melts                      115  
5 Sausage Biscuit (Regular Biscuit)    105.
```

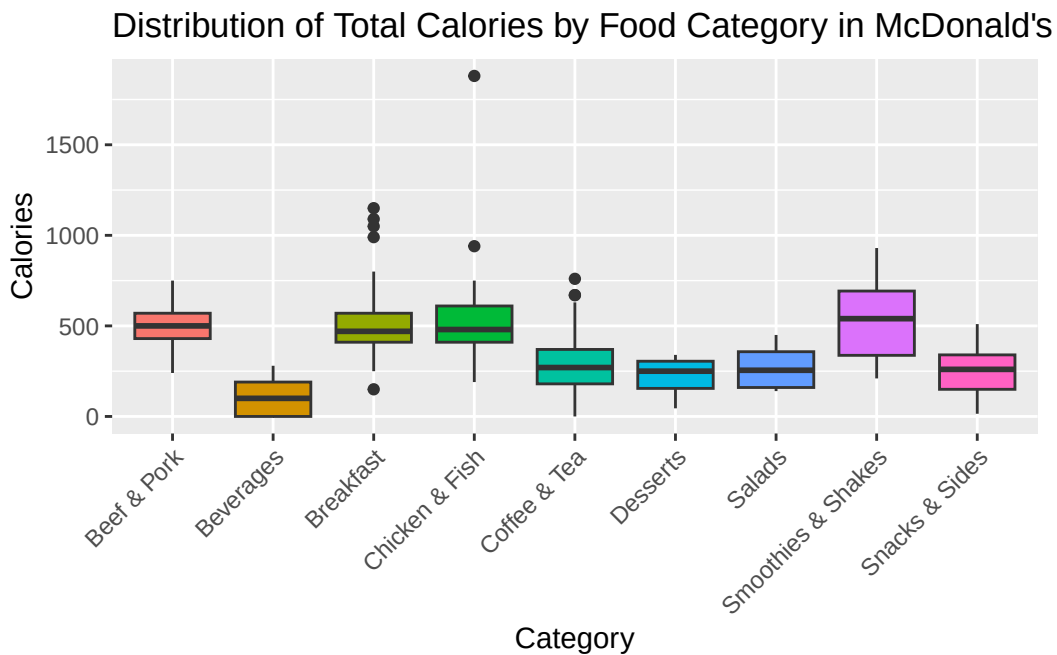
- c. Chicken McNuggets highest total calories. Chocolate Chip Cookie had highest calories per ounce. Not surprised given amount of breading on McNuggets and amount of sugar/butter in a cookie.

Exercise 5

```
mcd_cal <- mcd_cal |>
  mutate(prop_cal_fat = Fat*9/Calories)
```

Exercise 6

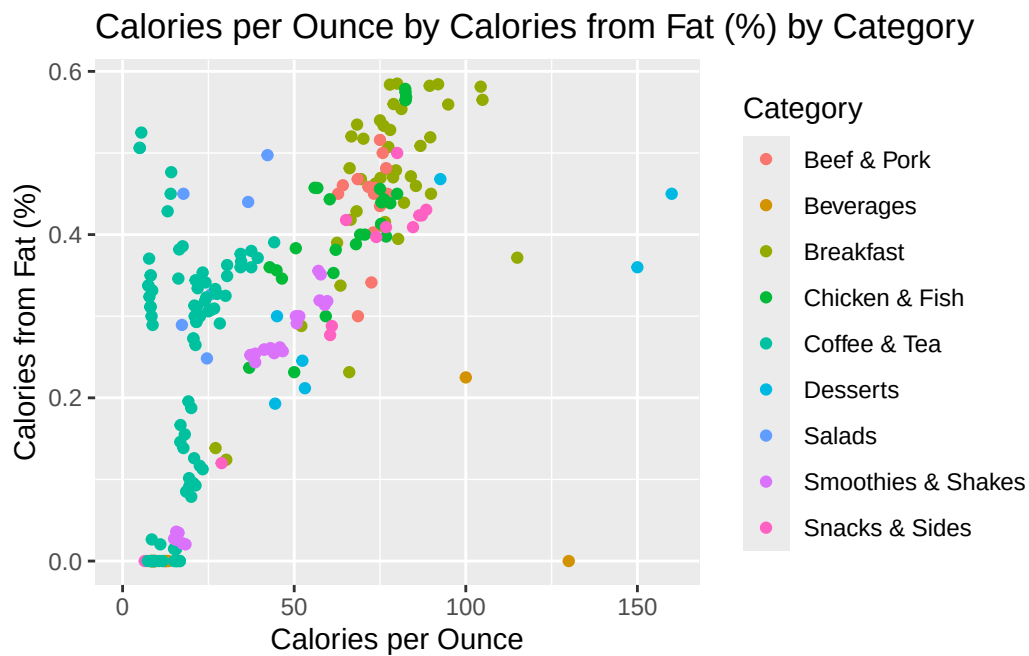
```
ggplot(data = mcd,
       mapping = aes(y = Calories,
                     x = Category,
                     fill = Category)) +
  geom_boxplot() +
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Distribution of Total Calories by Food Category in McDonald's Menu")
```



Exercise 7

```
ggplot(data = mcd_cal,
       mapping = aes(x = cal_oz,
                     y = prop_cal_fat,
                     color = Category)) +
  geom_point() +
  labs(title = "Calories per Ounce by Calories from Fat (%) by Category",
       x = "Calories per Ounce",
       y = "Calories from Fat (%)")
```

Warning: Removed 16 rows containing missing values or values outside the scale range (`geom\_point()`).



Breakfast has high calories per ounce and high calories from fat.

Coffee and tea has lower calories per ounce but higher calories from fat (somewhat).